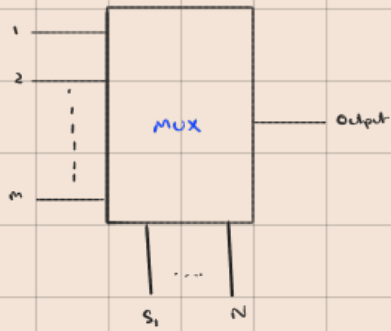
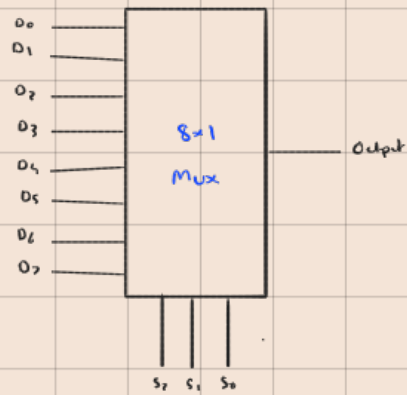
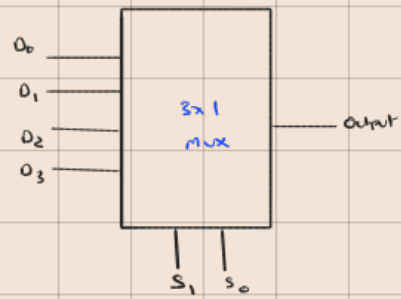


Multiplexer

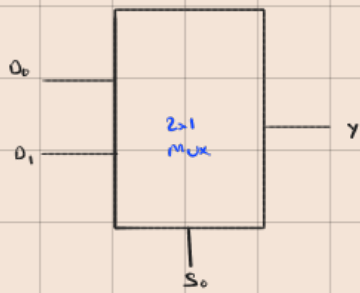


$$M = 2^N$$

$$N = \log_2(M)$$



2x1 Multiplexer



D_1	D_0	S_0	Y	Y
0	0	0	0	D_0
0	0	1	0	D_1
0	1	0	1	D_0
0	1	1	0	D_1
1	0	0	0	D_0
1	0	1	1	D_1
1	1	0	1	D_0
1	1	1	0	D_1

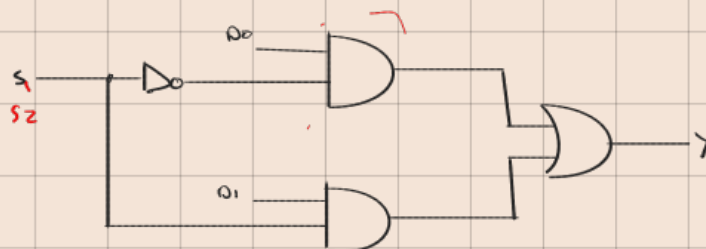
or

S_0	Y
0	D_0
1	D_1

$$Y = \overline{S_0} D_0 + S_0 D_1$$

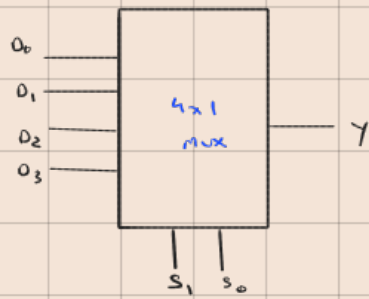
$S_0 D_1$

$$Y = \overline{S_0} D_0 + S_0 D_1$$



This is the same for all, just add more and gates.

4x1 multiplexer

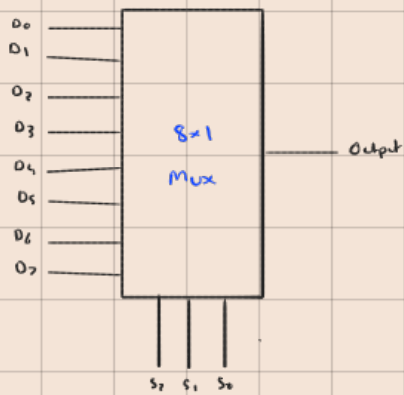


S_1	S_0	Y
0	0	D_0
0	1	D_1
1	0	D_2
1	1	D_3

$$Y = \bar{S}_1 \bar{S}_0 D_0 + \bar{S}_1 S_0 D_1 + S_1 \bar{S}_0 D_2 + S_1 S_0 D_3$$

$$Y = \bar{S}_1 \bar{S}_0 D_0 + \bar{S}_1 S_0 D_1 + S_1 \bar{S}_0 D_2 + S_1 S_0 D_3$$

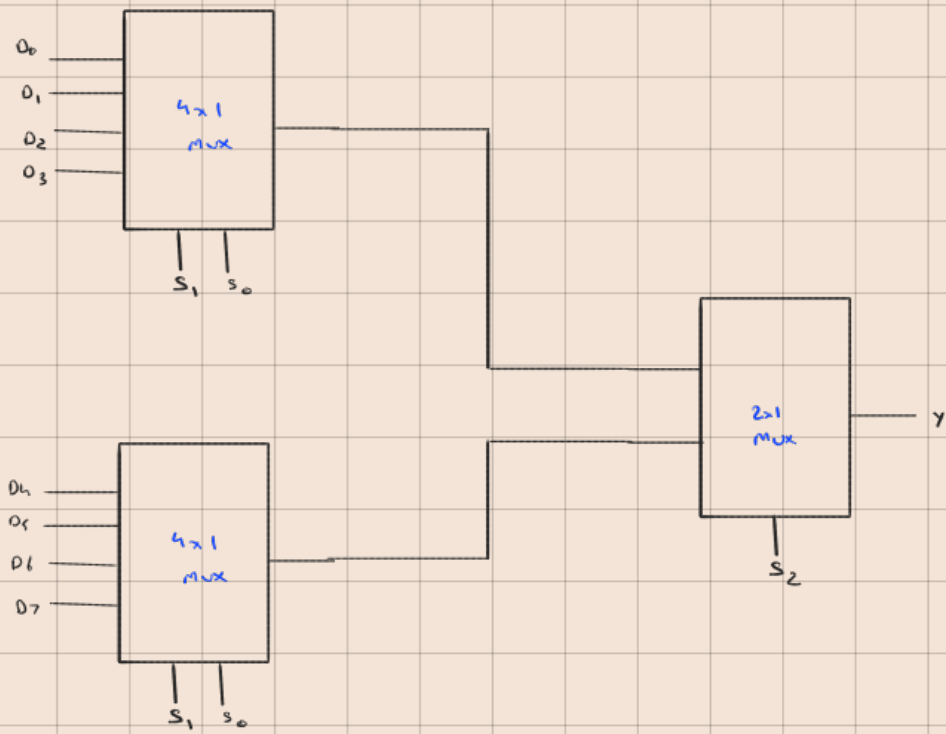
8x1 multiplexer



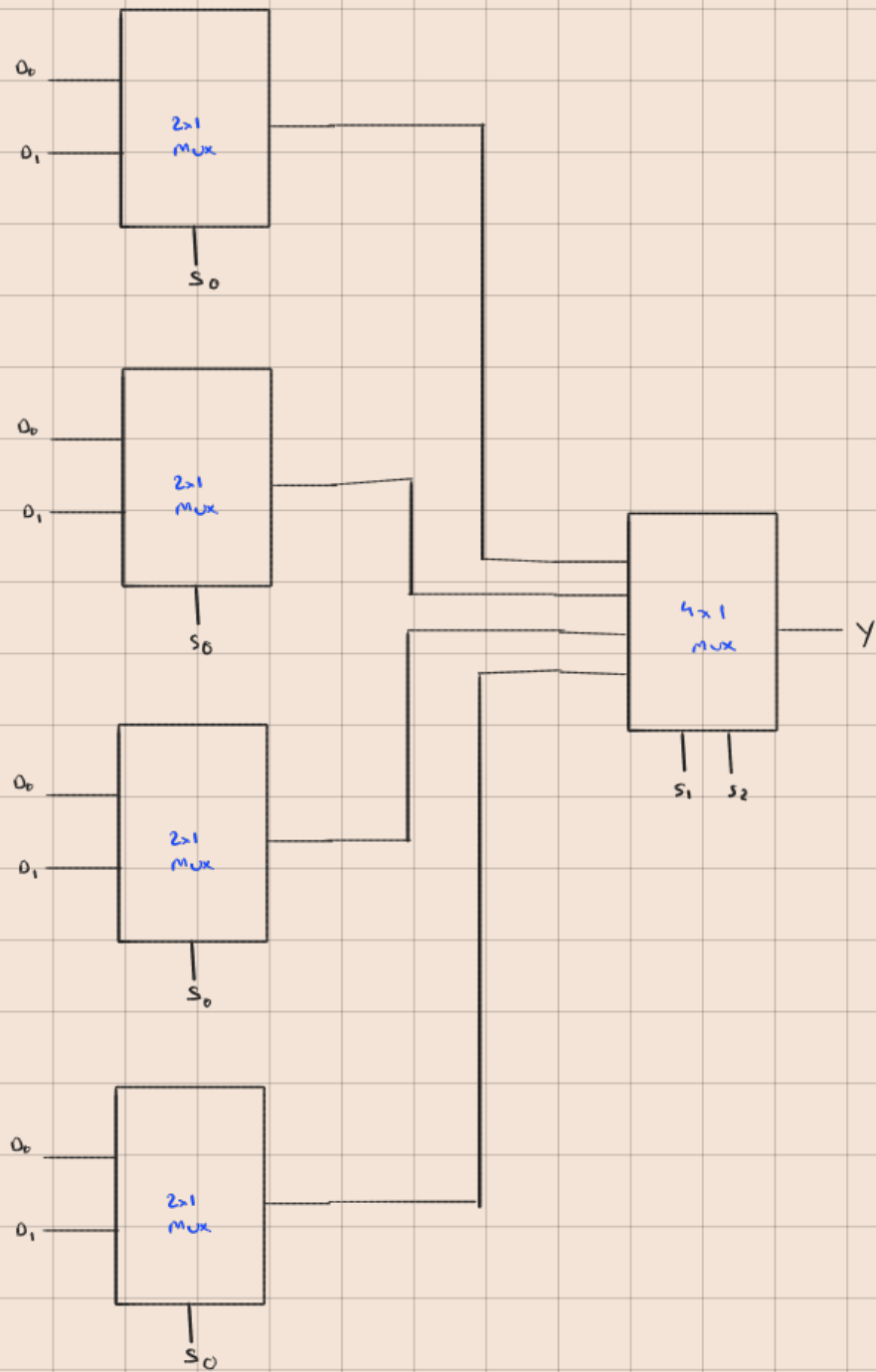
S_2	S_1	S_0	Y
0	0	0	D_0
0	0	1	D_1
0	1	0	D_2
0	1	1	D_3
1	0	0	D_4
1	0	1	D_5
1	1	0	D_6
1	1	1	D_7

Inputs = 2^{number of signals}

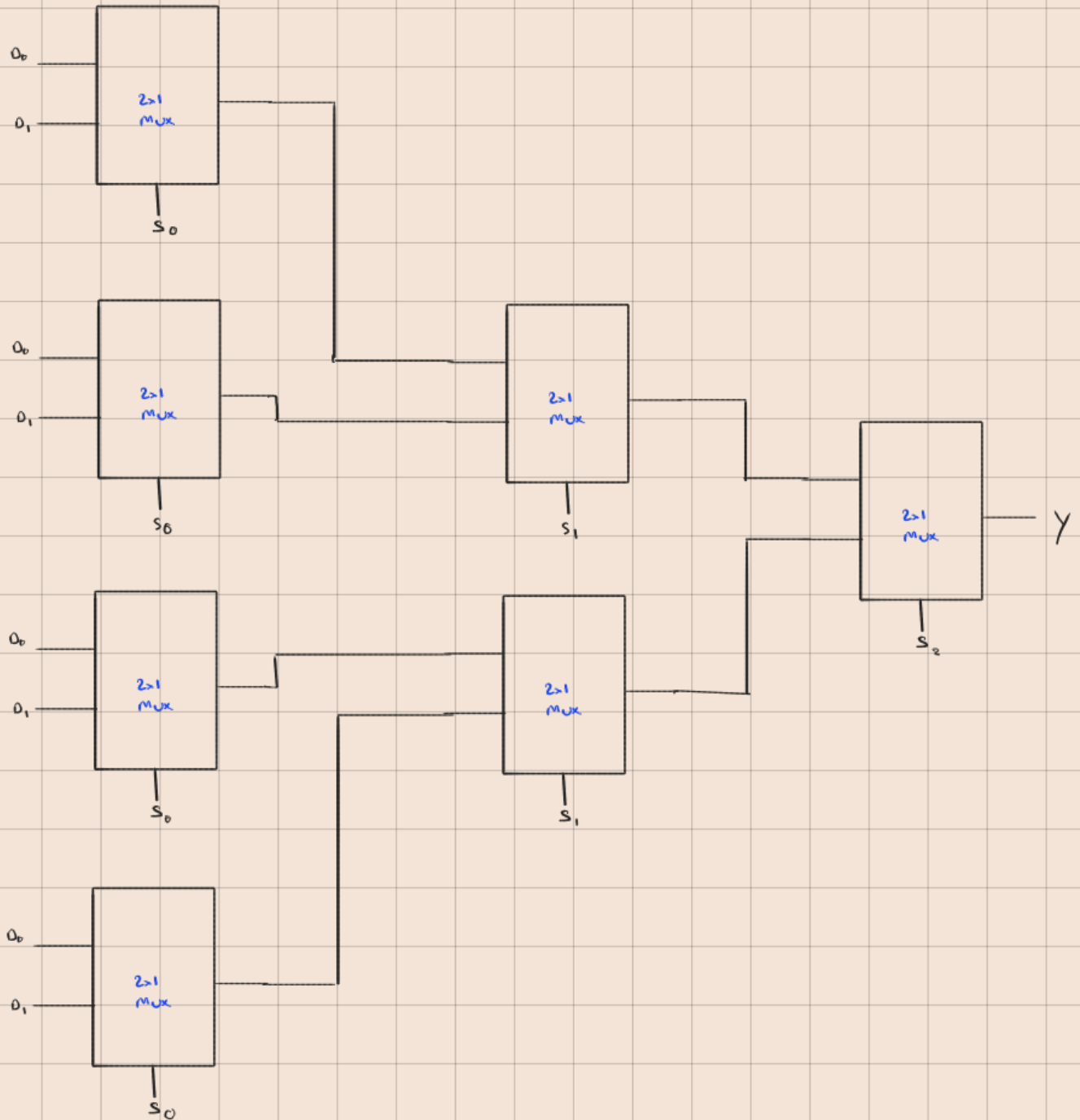
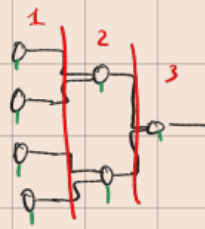
8x1 Mux Using 4x1₃ 2



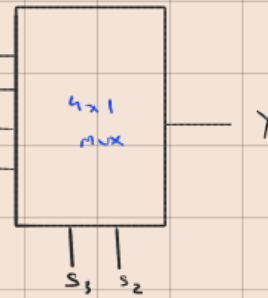
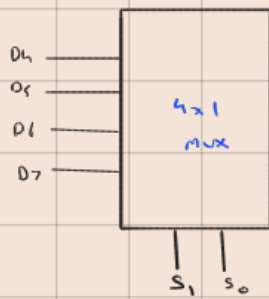
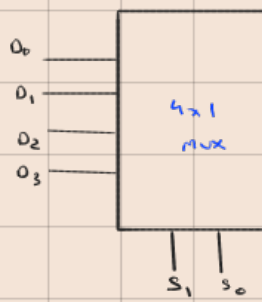
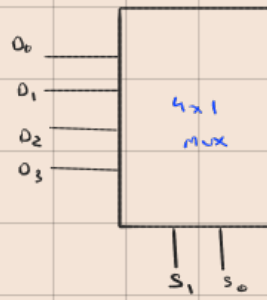
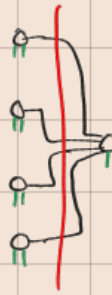
8x1 using 2x1
3



8x1 using 2x1 only

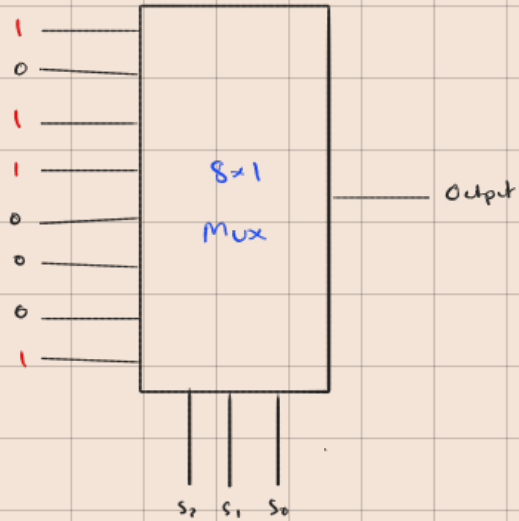


16x1 Using 4x1



Implementation of Boolean Logic

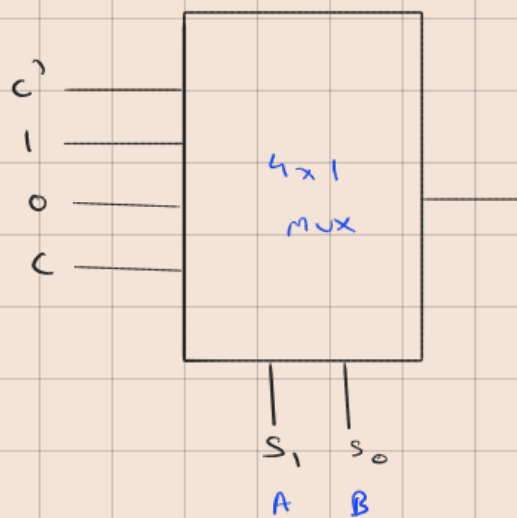
A	B	C	F	
0	0	0	1	D_0
0	0	1	0	D_1
0	1	0	1	D_2
0	1	1	1	D_3
1	0	0	0	D_4
1	0	1	0	D_5
1	1	0	0	D_6
1	1	1	1	D_7



Doing the same shift but with VEM

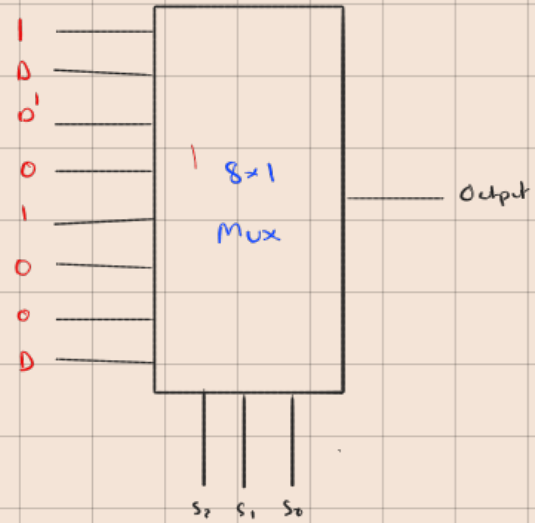
- n variables can be implemented using a mux with $n-1$ selection lines
- In earlier case we had 3 variables so we can use 4-1 mux which got 2 selection lines.

A	B	C	F	
0	0	0	1	C'
0	0	1	0	
0	1	0	1	1
0	1	1	1	
1	0	0	0	0
1	0	1	0	
1	1	0	0	C
1	1	1	1	



$$F(A,B,C,D) = \sum (0, 1, 3, 4, 8, 9, 15)$$

A	B	C	D	F
0	0	0	0	1
0	0	0	1	1
0	0	1	0	0
0	0	1	1	1
0	1	0	0	1
0	1	0	1	0
0	1	1	0	0
0	1	1	1	0
1	0	0	0	1
1	0	0	1	1
1	0	1	0	0
1	0	1	1	0
1	1	0	0	0
1	1	0	1	0
1	1	1	0	0
1	1	1	1	1



Full Adder Implementation using Mux

Full Adder-

A	B	Cin	Sum	Cout
0	0	0	0	0
0	0	1	1	0
0	1	0	1	0
0	1	1	0	1
1	0	0	1	0
1	0	1	0	1
1	1	0	0	1
1	1	1	1	1

A

B

Cin

